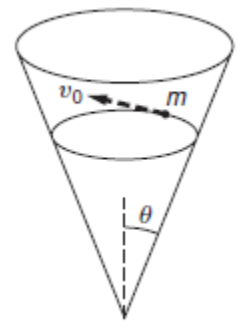
PHY101: Introduction to Physics-1

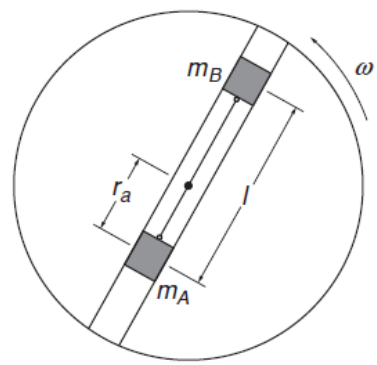
Monsoon, 2017

Tutorial-8

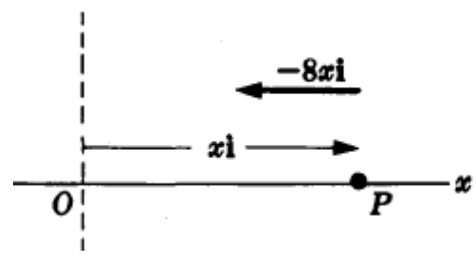
1. A particle of mass m slides without friction on the inside of a cone. The axis of the cone is vertical, and gravity is directed downward. The apex half-angle of the cone is θ, as shown. The path of the particle happens to be a circle in a horizontal plane. The speed of the particle is v_0. Draw a force diagram and find the radius of the circular path in terms of v_0, g, and θ.



2. A disk rotates with constant angular velocity ω, as shown. Two masses, m_A and m_B, slide without friction in a groove passing through the center of the disk. They are connected by a light string of length l, and are initially held in position by a catch, with mass m_A at distance r_A from the center. Neglect gravity. At t = 0 the catch is removed and the masses are free to slide. Find $\ddot{r}_A$ immediately after the catch is removed, in terms of m_A, m_B, l, r_A, and ω.



3. A particle P of mass 2 unit moves along x-axis, as shown in figure, attracted toward origin O by a force whose magnitude is numerically equal to 8x. If it is initially at rest at x=20 unit, find
(a) the differential equation of motion,
(b) the position of the particle at any time,
(c) the speed and velocity of the particle at any time, and
(d) the amplitude, period and frequency of the vibration.



4. A small metal sphere of radius 0.5 mm has mass 0.001 kg is dropped into water of which the viscosity is 0.36 Ns/m^2. What is the terminal velocity at which it falls?